



Situational Awareness of a Transmission Line

why we need V , I , P and Q , not the current alone

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1 Setup and line-only constants

The line is modelled in the distributed-parameter form developed in the parent technical document. The local terminal voltage is held balanced and positive sequence at the nominal line-to-neutral magnitude $V_s = V_{LL}/\sqrt{3}$, with phase ordering $\mathbf{u} = (1, a^2, a)^\top$ and $a = e^{j2\pi/3}$. The sending-end three-phase complex power and the line length are the two free parameters of the present analysis, $S = P + jQ \in \mathbb{C}$ and $\ell \in \mathbb{R}_+$.

The line geometry enters the equations only through the matrix exponential of the state-space generator, $\mathbf{M}(\ell) = e^{\mathbf{F}\ell}$, with $\mathbf{F} = [\mathbf{0}, -\mathbf{Z}; -\mathbf{Y}, \mathbf{0}]$ assembled from the per-unit-length series impedance and shunt admittance matrices. The upper blocks of $\mathbf{M}(\ell)$ are the ABCD operators $\mathbf{A}(\ell)$ and $\mathbf{B}(\ell)$, both of which are independent of S and V_s . From these blocks four *line-only* quantities are formed, all of which will be repeatedly invoked below:

$$\mathbf{p}(\ell) = \mathbf{B}(\ell)^{-1} \mathbf{u}, \quad \mathbf{q}(\ell) = \mathbf{B}(\ell)^{-1} \mathbf{A}(\ell) \mathbf{u}, \quad (1)$$

$$\alpha(\ell) = \mathbf{u} \cdot \overline{\mathbf{p}(\ell)}, \quad \beta(\ell) = \mathbf{u} \cdot \overline{\mathbf{q}(\ell)}, \quad (2)$$

where the dot product is the ordinary, non-Hermitian component sum and the overline denotes complex conjugation. The vectors $\mathbf{p}, \mathbf{q} \in \mathbb{C}^3$ carry the phase index $k \in \{A, B, C\}$ through their components p_k and q_k ; the scalars $\alpha, \beta \in \mathbb{C}$ do not. Every quantity in ((1))–((2)) depends on ℓ only and is computed once for a given tower geometry by exponentiating $\mathbf{F}$ at the requested length.

The two-point boundary problem solved in the parent document delivers the receiving-end complex unknown

$$W(\ell, S) = \frac{S + V_s^2 \beta(\ell)}{V_s \alpha(\ell)}, \quad (3)$$

and the sending-end phase currents follow from $\mathbf{I}_s = \mathbf{B}^{-1}(W^* \mathbf{u} - V_s \mathbf{A} \mathbf{u})$. Both relations are recalled here only to fix notation; the rest of the present note works directly in (S, ℓ) .

2 The phase current as an affine functional of S

Substituting Equation (3) into the sending-current relation $\mathbf{I}_s = \mathbf{B}^{-1}(W^* \mathbf{u} - V_s \mathbf{A} \mathbf{u})$ and reading off the k -th component yields, after a straightforward algebraic rearrangement,

$$I_k(S, \ell) = a_k(\ell) S^* + b_k(\ell), \quad k \in \{A, B, C\}, \quad (4)$$

where the two complex coefficients depend on the line only:

$$a_k(\ell) = \frac{p_k(\ell)}{V_s \alpha(\ell)}, \quad b_k(\ell) = \frac{V_s [p_k(\ell) \overline{\beta(\ell)} - \overline{\alpha(\ell)} q_k(\ell)]}{\alpha(\ell)}. \quad (5)$$

Each phase current is therefore an affine functional of the complex apparent power: $a_k(\ell) \in \mathbb{C}$

is the gain on the loading term S^* , and $b_k(\ell) \in \mathbb{C}$ is the no-load offset that the line delivers at $S = 0$. Both coefficients are produced once per length from Equation (1) and Equation (2), and the dependence on S is fully linear.

The affine form Equation (4) is exact within the model assumptions of the parent document (symmetric local-bus voltage, balanced remote-bus magnitude, lumped end-to-end $M(\ell)$) and absorbs every detail of the non-circulant per-unit-length matrices into the six complex line constants $\{a_A, a_B, a_C, b_A, b_B, b_C\}(\ell)$.

3 Modulus of the phase current as a quadratic form in (P, Q)

Taking modulus squared of Equation (4) and expanding,

$$|I_k(S, \ell)|^2 = |a_k(\ell)|^2 |S|^2 + 2 \operatorname{Re}[a_k(\ell) \overline{b_k(\ell)} S^*] + |b_k(\ell)|^2. \quad (6)$$

Splitting $S = P + jQ$ and using $\operatorname{Re}(zS^*) = \operatorname{Re}(z)P + \operatorname{Im}(z)Q$ gives the equivalent real-valued form

$$|I_k(P, Q, \ell)|^2 = \xi_k(\ell) (P^2 + Q^2) + 2 \operatorname{Re}[\eta_k(\ell)] P + 2 \operatorname{Im}[\eta_k(\ell)] Q + \zeta_k(\ell), \quad (7)$$

where the three line-only scalars per phase are

$$\xi_k(\ell) = |a_k(\ell)|^2, \quad \eta_k(\ell) = a_k(\ell) \overline{b_k(\ell)}, \quad \zeta_k(\ell) = |b_k(\ell)|^2. \quad (8)$$

Among these, ξ_k and ζ_k are non-negative reals and η_k is complex. They are derived once from Equation (5) for a given length.

Equation (7) states that the modulus squared of every phase current is a quadratic polynomial in the active and reactive power, with no cross product PQ , with the coefficient of $(P^2 + Q^2)$ equal across the three phases up to ξ_k , and with the linear part fully captured by the two real coordinates of the complex constant η_k . The asymmetry of the line lives entirely in the spread of $\{\xi_k, \eta_k, \zeta_k\}$ across $k \in \{A, B, C\}$.

The phase current modulus itself is the square root:

$$|I_k(P, Q, \ell)| = \sqrt{\xi_k(\ell) (P^2 + Q^2) + 2 \operatorname{Re}[\eta_k(\ell)] P + 2 \operatorname{Im}[\eta_k(\ell)] Q + \zeta_k(\ell)}. \quad (9)$$

For fixed ℓ the level sets of Equation (9) in the (P, Q) plane are circles centred at $(-\operatorname{Re}[\eta_k]/\xi_k, -\operatorname{Im}[\eta_k]/\xi_k)$; the displacement of those three centres is one geometric reading of the phase-current unbalance.

4 Peak-to-peak modular spread $\Delta_{\max}(S, \ell)$

The largest modular deviation between phases at the local terminal is the peak-to-peak spread of the three magnitudes Equation (9). With the line-only triplet $(\xi_k(\ell), \eta_k(\ell), \zeta_k(\ell))$ defined in Equation (8),

$$\Delta_{\max}(S, \ell) = \max_{k \in \{A, B, C\}} \sqrt{\xi_k |S|^2 + 2 \operatorname{Re}[\eta_k] P + 2 \operatorname{Im}[\eta_k] Q + \zeta_k} - \min_{k \in \{A, B, C\}} \sqrt{\xi_k |S|^2 + 2 \operatorname{Re}[\eta_k] P + 2 \operatorname{Im}[\eta_k] Q + \zeta_k}. \quad (10)$$

The two extrema are taken over the same scalar expression under the radical, evaluated with the per-phase constants (ξ_A, η_A, ζ_A) , (ξ_B, η_B, ζ_B) , and (ξ_C, η_C, ζ_C) . The whole right-hand side depends on ℓ through the nine line-only scalars and on the operating point through $(P, Q) = (\operatorname{Re} S, \operatorname{Im} S)$. No other quantity enters.

A useful corollary expresses the pairwise modulus difference in closed form. Subtracting Equation (7) for two phases i, j removes the constant of V_s and yields

$$|I_i|^2 - |I_j|^2 = (\xi_i - \xi_j) |S|^2 + 2 \operatorname{Re}[\eta_i - \eta_j] P + 2 \operatorname{Im}[\eta_i - \eta_j] Q + (\zeta_i - \zeta_j). \quad (11)$$

which is a pure scalar polynomial of degree two in (P, Q) , linear in the differences of the line constants only. The peak-to-peak spread Equation (10) is then equivalently the largest absolute value of $||I_i| - |I_j||$ over the three pairs $\{AB, BC, CA\}$.

4.1 Two operational corollaries

Light-loading asymptote. For small $|S|$ the quadratic term in $|S|^2$ is dominated by the linear and constant parts, and Equation (9) reduces to

$$|I_k(S, \ell)| \xrightarrow{|S| \rightarrow 0} \sqrt{\zeta_k(\ell)} + \frac{\operatorname{Re}[\eta_k(\ell)] P + \operatorname{Im}[\eta_k(\ell)] Q}{\sqrt{\zeta_k(\ell)}} + O(|S|^2). \quad (12)$$

At true no load ($S = 0$) the per-phase magnitude is exactly $\sqrt{\zeta_k(\ell)} = |b_k(\ell)|$, the line-charging current of each phase. The peak-to-peak spread then collapses to $\max_k |b_k(\ell)| - \min_k |b_k(\ell)|$, which depends on the line only.

Heavy-loading asymptote. For $|S|$ large the quadratic term in $|S|^2$ dominates and

$$|I_k(S, \ell)| \xrightarrow{|S| \rightarrow \infty} \sqrt{\xi_k(\ell)} |S| + \frac{\operatorname{Re}[\eta_k(\ell)] P + \operatorname{Im}[\eta_k(\ell)] Q}{\sqrt{\xi_k(\ell)} |S|} + O(|S|^{-3}). \quad (13)$$

The relative spread, that is the absolute spread divided by the average magnitude, then approaches the spread of $\sqrt{\xi_k(\ell)}$ across the three phases. Because $|a_k| = |p_k|/(V_s |\alpha|)$ is set

by the moduli of the phase-projected $B^{-1}\mathbf{u}$, the heavy-load relative unbalance is controlled purely by the asymmetry of $p(\ell)$ across the three phases.

5 Numerical example: 110 kV asymmetric single circuit

The line of the parent worked example is reused. A single conductor per phase of outer diameter 17.1 mm and DC resistance $0.1897 \Omega/\text{km}$ is placed at $(x, y)_A = (-4, 20)$, $(x, y)_B = (0, 20)$ and $(x, y)_C = (+4, 23)$ m above a perfectly conducting earth plane with one overhead ground wire of diameter 11.11 mm at $(0, 25)$ m. The frequency is 50 Hz and the soil resistivity is $1000 \Omega \text{ m}$.

Table 1 reports the nine line-only scalars of Equation (8) for two representative lengths, $\ell = 80 \text{ km}$ and $\ell = 200 \text{ km}$, with the apparent power scaled so that S enters Equation (7) in megavolt-amperes directly. The line-to-neutral magnitude is $V_s = 110/\sqrt{3} \text{ kV} \approx 63.509 \text{ kV}$.

ℓ [km]	phase k	ξ_k [A^2/MVA^2]	$\text{Re } \eta_k$ [A^2/MVA]	$\text{Im } \eta_k$ [A^2/MVA]	ζ_k [A^2]
80	A	25.583	-0.9093	-0.8940	0.0636
	B	30.616	+1.1681	+0.2119	0.0460
	C	26.760	-0.0704	+0.6697	0.0169
200	A	25.596	-2.2767	-2.2368	0.3980
	B	30.609	+2.9206	+0.5325	0.2879
	C	26.754	-0.1737	+1.6716	0.1056

Table 1: Line-only triplets (ξ_k, η_k, ζ_k) for the 110 kV example, with S in MVA. Units: ξ in A^2/MVA^2 , the real and imaginary parts of η in A^2/MVA , and ζ in A^2 .

The structural asymmetry between the three phases is visible directly in the table. The quadratic coefficient $\xi_B \approx 30.6 \text{ A}^2/\text{MVA}^2$ is roughly 20 % larger than ξ_A and ξ_C ; phase B also carries the largest positive $\text{Re } \eta$ and the smallest $\text{Im } \eta$, which combine to produce the largest current modulus at active loading and unity-leading reactive power. The no-load offsets $\sqrt{\zeta_k}$ are 0.25, 0.21 and 0.13 A at 80 km, growing to 0.63, 0.54 and 0.32 A at 200 km because of the quadratic dependence of the line charging current on length.

Evaluating Equation (9) at $S = 80 + j10 \text{ MVA}$ through the table gives $|I_A| = 407.59 \text{ A}$, $|I_B| = 446.31 \text{ A}$ and $|I_C| = 417.06 \text{ A}$ at 80 km, with peak-to-peak spread

$$\Delta_{\max}(80 + j10 \text{ MVA}, 80 \text{ km}) = 446.31 \text{ A} - 407.59 \text{ A} \approx 38.73 \text{ A}, \quad (14)$$

and analogously 39.19 A at 200 km, in exact agreement with the direct propagation of the boundary problem. The example confirms that Equation (10) is a faithful and operationally usable closed form for the peak-to-peak modular spread of the phase currents at the local terminal, as a function of the three-phase complex apparent power and the line length only.

Table 2 consolidates the complete per-phase state at the anchor $S = 80 + j10$ MVA, $\ell = 25$ km that is referenced in the conclusion of section 7.

Phase	$ I_k $ [A]	$\arg I_k$ [°]	P_k [MW]	Q_k [MVA _r]
<i>A</i>	407.71	−3.77	25.84	1.70
<i>B</i>	446.17	−126.92	28.13	3.42
<i>C</i>	417.07	+109.38	26.03	4.88
Sum / mean	mean 423.65	—	80.00	10.00

Table 2: Per-phase electrical state at the anchor $S = 80 + j10$ MVA, $\ell = 25$ km. The receiving-end line-to-neutral voltage is $V_r = 61.08$ kV (96.17 % of V_s) at a transmission angle of $\delta = -3.90^\circ$. The peak-to-peak spread is $\Delta_{\max} = 38.47$ A (9.08 % of the mean), the positive-sequence current is $|I_+| = 423.16$ A, the negative-sequence current is $|I_-| = 25.95$ A, and the current unbalance factor is $\text{IUF} = |I_-|/|I_+| = 6.13\%$. Per-phase P_k and Q_k are computed as $\text{Re}\{V_s u_k I_k^*\}$ and $\text{Im}\{V_s u_k I_k^*\}$.

6 Parametric sweeps of $\Delta_{\max}$

Equation (10) is evaluated on two single-variable sweeps of the 110 kV asymmetric configuration of section 5. Figure 1 fixes the line length at $\ell = 25$ km and sweeps $|S|$ from no load up to 200 MVA along the direction of the example operating point, that is $\arg(S) = \arg(80 + j10 \text{ MVA}) \approx 7.13^\circ$. Figure 2 fixes the apparent power at the example value $S = 80 + j10$ MVA and sweeps the line length from 1 km up to 100 km. The filled marker on each curve is the common anchor at $S = 80 + j10$ MVA and $\ell = 25$ km.

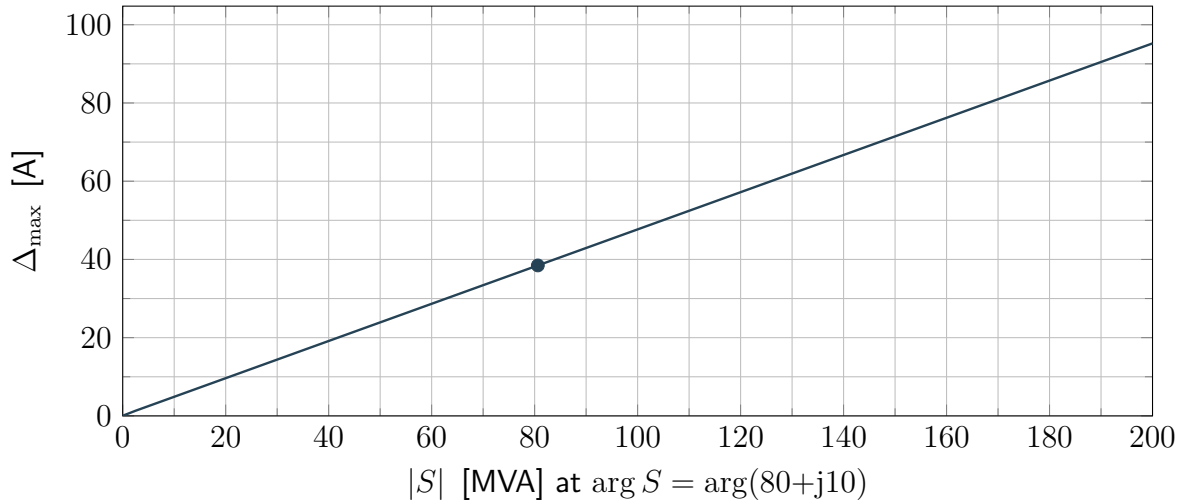


Figure 1: Peak-to-peak modular spread $\Delta_{\max}$ at the local terminal versus apparent-power magnitude $|S|$, with the line length held at $\ell = 25$ km and the loading direction held at the operating power factor of the example. The filled marker is the anchor at $S = 80 + j10$ MVA.

Figure 1 confirms the heavy-load asymptote of Equation (13): the spread is linear in $|S|$ across the plotted range, with a slope set by the spread of $\sqrt{\xi_k(\ell)}$ across the three phases.

The line passes through the origin to within the no-load offset $\max_k |b_k| - \min_k |b_k| < 0.05 \text{ A}$ that becomes visible only at $|S| \lesssim 0.5 \text{ MV A}$ and is not separable from zero at the scale of the figure.

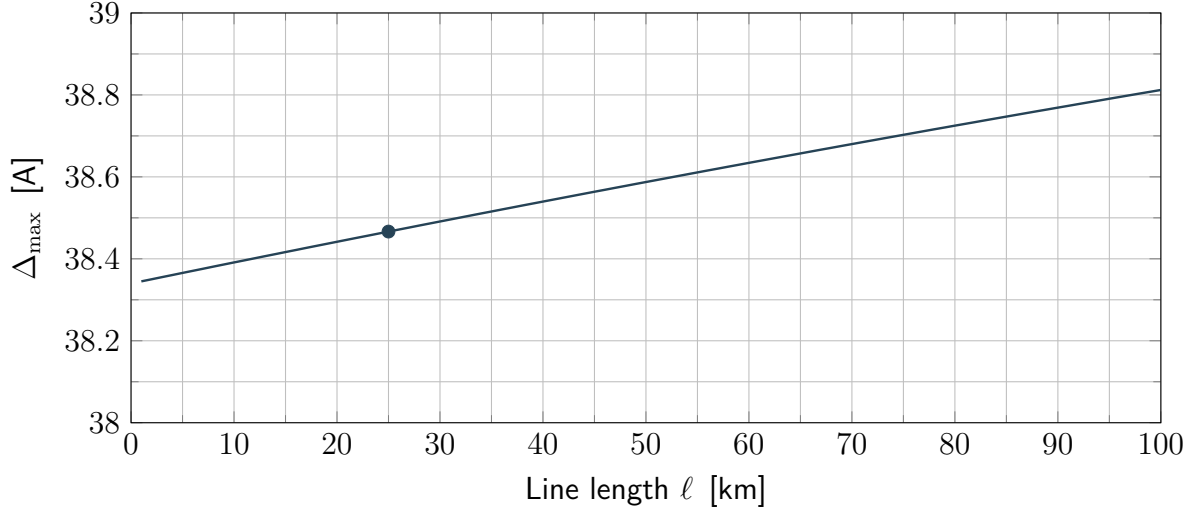


Figure 2: Peak-to-peak modular spread $\Delta_{\max}$ at the local terminal versus line length ℓ , with the apparent power held at the example value $S = 80 + j10 \text{ MV A}$. The filled marker is the anchor at $\ell = 25 \text{ km}$.

Figure 2 shows that for fixed S the spread is remarkably flat across the entire span of practical 110 kV corridor lengths. The variation across $\ell \in [1, 100] \text{ km}$ is bounded by 0.2 A around a base value of roughly 38.4 A , which is at the resolution edge of substation ammeters and consistent with the observation that, for fixed loading, the unbalance pattern is set by the asymmetry of the per-unit-length matrices and not by the length. The length enters $\Delta_{\max}$ through the ABCD blocks of $M(\ell) = e^{F\ell}$, but for the geometry under study the resulting variation of $\sqrt{\xi_k(\ell)}$ across the three phases is essentially invariant in ℓ ; the asymptote of Equation (13) is attained almost immediately and stays there as the line is lengthened.

7 Conclusion

The closed-form expression for the peak-to-peak modular spread $\Delta_{\max}(S, \ell)$ delivers a single operational reading: the actual electrical state of a terminal cannot be determined from the phase current alone, whether read as a single-phase modulus or as the full three-phase set. The reason is structural. The line is geometrically not circulant, and under balanced positive-sequence voltage at both terminals the natural asymmetry of the per-unit-length matrices has no degree of freedom left in the voltage state; it is forced into the current and into the per-phase split of P and Q . The current is therefore unbalanced by construction, set by the tower geometry rather than by a fault or an asymmetric excitation. At the anchor of section 5 this materialises as a 9.08% peak-to-peak spread and a 6.13% negative-sequence ratio, both produced entirely by geometry.

In any actual deployment the voltages at the two ends are not perfectly balanced either, so the asymmetry redistributes across all four signals – voltage, current, active and reactive power per phase – in proportions that depend on the loading direction and on the stiffness of each bus. A current-only reading, even on all three phases, cannot recover that distribution. The actual state of the terminal is the four-signal package (V, I, P, Q) per phase, with phase angles, measured at both ends.